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YOLOv8-Cherry: A Gated Multi-Task Collaborative Network for Cherry Detection and Grasping Keypoint Localization in Complex Orchard Environments (April 2024)

Qiaochu Li [1,3], Junyuan Zhang [1], Yingnan Zhou [1], Chen Zhang [1], Zhengzhou Li [2], Jianguo Chen [1,3]

[1]Engineering and Technical College of Chengdu University of Technology, Leshan, 614000, China
[2]College of Microelectronics and Communication Engineering, Chongqing University, Chongqing, 400044, China
[3]Intelligent Technology Application Research Center, Leshan, 614000, China

Corresponding author: Zhengzhou Li (e-mail: lizhengzhou@cqu.edu.cn).

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ABSTRACT In complex orchard environments, achieving multi-task synergy between cherry maturity detection and grasping keypoint localization poses a core challenge for agricultural robotic vision systems. Among the YOLO series, YOLOv8n achieves superior detection accuracy with comparable parameter counts, benefiting from its anchor-free paradigm and improved C2f module, while also offering faster inference and lower memory consumption on edge devices. With its compact size of approximately 6 MB and modular head design, it strikes an ideal balance among accuracy, efficiency, and scalability. However, in multi-task scenarios, it treats detection and keypoint localization as independent tasks with decoupled head structures, lacking explicit cross-task interaction, which consequently leads to gradient conflicts and insufficient feature utilization, making it difficult to achieve an ideal performance balance between the two tasks. To address the identified limitations, this paper proposes YOLOv8-Cherry, a multi-level task-cooperative enhancement network built upon YOLOv8n. Critically, unlike prior YOLO-based multi-task approaches that rely solely on parameter sharing or isolated task-specific branches, the primary novelty of our method lies not in any single component, but in the synergistic combination architecture of three collaboratively designed modules, orchestrated to enable bidirectional information flow between detection and keypoint tasks. Specifically: (1) a Dual-Task Synergistic Attention Module (DTSAM) introduces a learnable cross-task gating mechanism at the detection head level, dynamically filtering and exchanging complementary task-specific cues; (2) a Task-Balanced C2f Module (TBC2f) adaptively reweights multi-scale features during neck fusion to mitigate gradient conflicts and balance the representation strengths for both tasks; and (3) a Task Harmonic Attention Module (THAM) coordinates attention responses in shallow feature layers, ensuring that early-stage spatial details are preserved for keypoint localization without sacrificing semantic cues for detection. Among these, the gating mechanism in DTSAM serves as the pivotal enabler of cross-task synergy, while TBC2f and THAM provide essential feature- and spatial-level balance—a hierarchical cooperative paradigm that existing multi-task YOLO designs have overlooked. Experiments on our self-built Cherry dataset demonstrate that YOLOv8-Cherry improves detection mAP@0.5 by 6.6% and keypoint mAP@0.5 by 6.9% over the baseline YOLOv8n, while maintaining the efficiency of single-stage inference without introducing significant computational overhead. Furthermore, deployment on the NVIDIA Jetson Orin Nano validates the model's favorable portability and real-time operation capability on low-cost embedded devices, confirming its practical viability for robotic harvesting systems.

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INDEX TERMS cherry picking, multi-task learning, keypoint detection, ripeness detection, YOLOv8, attention mechanism, agricultural robot vision, real-time detection.

I. INTRODUCTION

From an economic value perspective, cherry holds high commercial value, demonstrating strong growth momentum and significant economic potential in the global market. Cherry cultivation is economically highly viable, with its profitability evidenced by high net present value, internal rate of return, and benefit-cost ratio[1]. Cherries also have high nutritional value, as they are rich in important nutrients and bioactive compounds such as dietary fiber, polyphenolic compounds, carotenoids, vitamin C, and potassium[2]. However, the harvesting process faces significant technical challenges. Research indicates that the cherry harvesting window is extremely limited; for most varieties, the transition from full ripeness to over-ripeness takes only a few days. Over-ripe fruits rapidly soften and decay, thereby significantly reducing their economic value[3]. Currently, cherry harvesting relies primarily on manual labor, which often faces labor shortages during peak agricultural seasons. Furthermore, manual harvesting incurs high costs, and issues such as mechanical damage during picking and handling frequently occur, severely affecting fruit quality and commercial value.

Therefore, achieving accurate identification and automated picking of cherries has become an urgent demand for the development of intelligent agriculture. As the core technological foundation of visual intelligent picking systems, the accuracy, robustness, and processing speed of keypoint detection and object detection technologies directly affect the economic benefits of the cherry industry.

However, existing detection technologies still face multiple challenges: most models can only achieve basic fruit localization, but due to the short ripening period of cherries and significant variations in ripeness among fruits from the same batch, the lack of a ripeness pre-judgment mechanism can easily lead to the premature picking of immature fruits, thereby affecting overall product quality[4, 5]; Close contact between fruits often causes extensive occlusion of key feature points such as pedicels and fruit navels, leading to feature confusion and ambiguous localization. The small size of cherry fruits means that their keypoints occupy only a very small proportion of pixels in the image, especially at the pedicel end and the navel depression where morphological features are subtle, which directly reduces the detector's recall rate and localization accuracy; Complex field environmental factors, such as uneven illumination, leaf occlusion, branch shadows, and the presence of water droplets or dirt, can severely obscure or distort the apparent features of keypoints, making accurate recognition particularly difficult[6].

Current cherry detection mainly relies on three types of techniques. The first is traditional image processing methods, such as corner detection, template matching, or handcrafted feature extraction combined with classifiers. Although these methods are computationally efficient, their feature representation is limited. In real-world scenarios involving variations in fruit color, diverse fruit postures, and fluctuating illumination, their generalization ability is insufficient, making it difficult to reliably handle complex situations. The second type involves three-dimensional structured light or laser scanning solutions, which acquire depth information and geometric contours of the fruit surface, offering advantages in spatial localization. However, these methods involve high hardware costs and have limited capability in capturing surface visual features such as color and texture, thereby restricting their potential for application in multi-feature fusion detection. The third type is a detection framework based on deep learning. He et al.[7] developed a rice seedling detection model based on YOLOv5-Lite, achieving an mAP@0.5 of 81.9%. The model was deployed on an NVIDIA Jetson Nano, where the monitoring system attained 99.2% accuracy in seedling counting and 90.3% accuracy in counting missing seedlings, with a processing speed of 3.95 frames per second. Yang *et al.*[8] proposed a lightweight and fast strawberry growth state detection model based on YOLOv7-tiny. The model achieved a mAP@0.5 of 97.5% and was successfully deployed on desktop and Android devices, enabling real-time and efficient detection in resource-constrained environments. Gu et al.[9] proposed an improved model based on YOLOv8, achieving a detection accuracy of 97.63% for mango fruits and 94.5% for fruit stems. The model was deployed on an NVIDIA Jetson Orin Nano, enabling effective detection of both fruits and stems. Although the aforementioned models balance detection accuracy and deployment efficiency on devices, they lack adaptability, are easily affected by complex orchard backgrounds, and perform poorly on small and occluded targets.

To address the above issues, this paper proposes a multi-task collaborative enhancement network (YOLOv8-Cherry Network). Based on the YOLOv8-Pose framework, this network introduces three hierarchical collaborative modules: Dual-Task Balanced C2f Module (TBC2f), Task Harmonic Attention Module (THAM), and Dual-Task Synergistic Attention Module (DTSAM). These modules collectively form a comprehensive collaborative system that spans from feature extraction to feature enhancement, enabling dynamic balancing of feature representations between the detection task (classification and localization of ripe/semi-ripe cherries) and

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the keypoint estimation task (localization of five picking-related keypoints). The core innovation of the proposed method lies in:

- *This paper proposes a multi-level task collaborative enhancement network architecture based on YOLOv8-Pose, termed the YOLOv8-Cherry Network. Through the Dual-Task Balanced C2f Module (TBC2f), the Task Harmonic Attention Module (THAM), and the Dual-Task Synergistic Attention Module (DTSAM), it constructs a collaborative optimization framework that spans from the backbone to the detection heads, achieving adaptive feature interaction and dynamic balancing between detection and keypoint tasks. The model introduces a ripeness assessment mechanism prior to fruit picking, ensuring that only ripe cherries are harvested. In addition, this study validates the edge-side deployment of the model on an NVIDIA Jetson Orin Nano. After deployment, the model can operate in a resource-constrained edge environment while maintaining both detection accuracy and efficiency, thus meeting the practical requirements of real-time monitoring and keypoint localization for orchard cherries.*
- *We design a lightweight task-interaction gating network and a harmonic weight generation mechanism, which can dynamically adjust the task attention strategies based on the input semantics during feature fusion and attention allocation, effectively alleviating feature competition and conflict among multiple tasks, and enhancing the model's robustness under occlusion, small objects, and complex backgrounds.*
- *We construct a multi-task evaluation framework and a high-quality dataset (Cherry Dataset) specifically for cherry picking scenarios, covering cherry samples with varying ripeness levels, occlusion conditions, and complex environments. This dataset provides a reliable benchmark for fruit detection and keypoint localization tasks, and demonstrates that the proposed method significantly improves detection and localization accuracy while maintaining real-time performance.*

II. RELATED WORK

A. OBJECT DETECTION

As an important branch of computer vision, object detection aims to accurately locate and identify objects from images or videos. From early traditional methods based on handcrafted features to modern end-to-end models based on deep neural networks, this field has undergone decades of development and transformation. Among traditional methods, representative ones include the Viola-Jones algorithm[10] , HOG features[11] combined with SVM classifiers[12] ,and the SIFT algorithm[13]. The Viola-Jones algorithm utilizes Haar features and an AdaBoost cascade classifier, achieving fast speed in face detection. However, it has insufficient adaptability to pose variations and occlusion. The HOG combined with SVM method, based on histograms of oriented gradients, has certain robustness to local deformation and illumination changes and performs well in pedestrian detection. However, it suffers from high feature dimensionality and computational cost, and its generalization ability to occlusion and complex backgrounds is relatively weak. The SIFT algorithm achieves scale invariance through Gaussian pyramids and difference-of-Gaussian pyramids[14] , and ensures rotation invariance using orientation information, while also exhibiting good stability under illumination changes. However, it has high computational cost and low real-time performance, making it difficult to extract stable features from smooth or blurred images. Traditional methods work well in simple scenarios, but due to their limited feature generalization and low computational efficiency, they fail to handle complex scenes. With the development of deep learning, convolutional neural networks[15] are able to automatically learn features ranging from low-level edges to high-level semantics, achieving joint optimization of feature extraction and detection tasks in an end-to-end manner, thereby advancing object detection toward high robustness and applications in complex scenes. Deep learning-based object detection methods can be mainly categorized into two-stage methods, single-stage methods, anchor-free methods, and Transformer-based methods. Two-stage methods first generate candidate regions and then perform classification and location refinement on these candidate regions, achieving high accuracy but at a relatively slower speed. R-CNN[16] employs selective search to generate candidate boxes, uses convolutional networks for feature extraction, and applies SVM for classification. However, its pipeline is disjointed, training is complex, and storage overhead is high. Fast R-CNN[17] introduces feature sharing and RoI pooling, integrating feature extraction, classification, and regression into a single network, which significantly improves training efficiency. However, candidate box generation still relies on external algorithms and is not fully end-to-end. Faster R-CNN[18] proposes the Region Proposal Network (RPN), which integrates candidate box generation into the network and enables end-to-end training, significantly improving detection speed. However, its performance is affected by the preset anchor box sizes, and its adaptability to small targets and dense scenes remains insufficient. Single-stage methods directly predict categories and bounding boxes on the image, achieving faster speed and being more suitable for real-time detection. YOLOv1[19] treats detection as a regression problem and achieves global contextual awareness through grid division. However, it has weak capability in small object detection and limited localization accuracy. YOLOv7[20] further proposes an efficient layer aggregation network and a dynamic label assignment mechanism, leveraging multi-scale features to improve detection performance on small objects. However, false positives and missed detections under extreme scale variations, severe occlusion, and complex backgrounds still require improvement. Anchor-free methods discard preset

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anchor boxes and directly predict object keypoints or boundaries.

HDPNet[21] achieves high detection accuracy in low-light environments. However, its MDN (Mixture Density Network) module and IAFN (Illumination-Aware Feature Network) module incorporate Transformers, which increases the computational burden. FCOS[22] is based on a fully convolutional structure, regressing for each point on the feature map its distances to the four sides of the target, and introduces "centerness" to improve detection accuracy. It performs well in dense and small object detection. However, overlapping targets can lead to ambiguity in feature point assignment, which may cause false positives or duplicate predictions. Transformer-based methods utilize the self-attention mechanism to model global dependencies in images, achieving end-to-end detection in a sequential manner. DETR[23] is the first to introduce the Transformer into detection tasks, directly outputting detection results through an encoder-decoder structure without requiring anchor boxes or NMS (Non-Maximum Suppression), thereby simplifying the pipeline. However, it suffers from slow training convergence, weak capability in small object detection, and high computational complexity. The Swin Transformer[24] adopts a hierarchical design and a window-based self-attention mechanism, significantly reducing computational cost and being suitable for multi-scale vision tasks. However, its shifted window mechanism may limit global information interaction, and it still requires relatively high computational power.

Apart from network-structure modifications, another effective paradigm for boosting object detection performance concentrates on dataset construction, specialized annotation schemes and training-level optimization. For small-object detection under challenging imaging conditions, well-designed training data and training strategies can significantly mitigate feature degradation caused by distortion and low resolution. Representatively, E-YOLO to OMOD targets small moving-object detection in omnidirectional videos [25]. This work leverages transfer learning and constructs a carefully curated omnidirectional dataset, where key feature regions of moving objects are specially annotated to counteract resolution limitations. Although it achieves notable single-task detection improvements, its performance gain mainly comes from data curation and training strategies rather than intrinsic network-level multi-task interaction. This method only focuses on standalone object detection and lacks mechanisms to model and balance detection and key-point localization simultaneously. Beyond algorithmic advancements, the choice of deployment hardware is also critical for real-world applications. Sendo and Premachandra[26] demonstrated a wheelchair navigation system that uses YOLOv8 deployed on an embedded platform to detect Braille blocks in real time, achieving 97.3% detection accuracy in both indoor and outdoor environments. Their work highlights the importance of balancing detection performance with hardware constraints in mobile scenarios.

B. KEYPOINT-BASED DETECTION

Early keypoint detection mainly relied on traditional image processing methods, such as Harris corner detection[27], SIFT, SURF[28], FAST[29] and other feature descriptors, combined with sliding windows and classifiers such as SVM for detection. In corner detection-based algorithms, a fixed window is slid over the image in arbitrary directions. The presence of a corner is determined by comparing the degree of pixel intensity change within the window before and after sliding. Harris corner detection extracts corners directly based on grayscale images and achieves high detection accuracy for L-shaped corners. However, it employs Gaussian filtering[30] , which results in slow computational speed. Moreover, it does not possess scale invariance; when the image scale changes, the corners may also change accordingly. The SIFT algorithm identifies potential scale- and rotation-invariant keypoints using the difference-of-Gaussian function. At each candidate location, a refined model is fitted to determine the position and scale, and then one or more orientations are assigned to each keypoint based on local image gradients. Finally, within the neighborhood of each keypoint, gradients are measured on selected images; these gradients serve as the keypoint descriptor, allowing for significant shape deformation and illumination changes in the image. However, the SIFT algorithm involves a large amount of computation, resulting in low real-time performance. Meanwhile, when the image edges are relatively smooth or the image is blurred, the number of feature points that the algorithm can detect may decrease, making it difficult to extract stable features. The SURF algorithm first constructs an integral image, then builds a Hessian matrix to construct a Gaussian pyramid scale space. Next, it uses non-maximum suppression to preliminarily identify feature points, then precisely localizes the extreme points, and finally selects the main orientation of the feature points and constructs the SURF feature point descriptor. The SURF algorithm accelerates computation by constructing an integral image. Moreover, it achieves scale invariance by building a multi-scale space. However, the stage of determining the main orientation relies too heavily on the gradient directions of pixels in the local region, which may lead to inaccuracies in the found main orientation. Moreover, its image pyramid is not constructed densely enough, which may introduce scale errors. The FAST algorithm compares the intensity values of pixels within a certain neighborhood to that of the center point to determine whether a corner exists. It does not involve complex operations such as scale or gradient, resulting in fast detection speed. However, it lacks orientation and scale invariance.

In 2014, DeepPose[31] was proposed at the CVPR conference, marking the first framework to apply deep neural networks to human pose estimation and keypoint localization. This framework introduced a cascaded network structure that progressively refines the positions of keypoints through multiple stages, thereby improving detection accuracy. However, it still performs poorly in cases of keypoint overlap

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and occlusion. The Stacked Hourglass Network[32] is a fully convolutional network that performs multiple up-sampling and down-sampling operations, forming multi-scale feature representations and enhancing sensitivity to resolution in pose estimation. However, due to its repeated down-sampling and up-sampling, as well as the stacking of multiple modules, the computational cost is relatively high. The Cascaded Pyramid Network (CPN)[33] adopts a two-stage cascade structure. Its GlobalNet captures the overall structure of all keypoints and performs initial keypoint prediction. Its RefineNet then processes difficult keypoints to improve detection accuracy. This structure achieves high detection accuracy and robustness. However, its cascaded computation increases computational overhead, leading to insufficient real-time performance and making it difficult to deploy on edge devices. AlphaPose[34] first crops out individual human regions based on object detection bounding boxes, then uses a symmetric spatial transformer network to perform pose normalization and correction. Next, it employs a stacked hourglass network to predict keypoints on the corrected image. Finally, it applies pose-based non-maximum suppression to remove and merge duplicate or overlapping poses in multi-person scenarios, outputting the final multi-person poses. AlphaPose achieves high accuracy in multi-person pose estimation tasks. However, during inference, it requires storing cropped images and feature maps for multiple individuals, which consumes a significant amount of memory. Moreover, because it must first perform detection before pose estimation, its network structure is complex, resulting in limited real-time performance. HRNet[35] connects multiple sub-networks of different resolutions in parallel. By repeatedly performing cross-resolution information fusion, the high-resolution feature maps continuously receive semantic information from low-resolution features. Finally, the network directly predicts keypoint heatmaps from the highest-resolution feature maps, achieving keypoint localization. Due to the need to maintain high-resolution features and fuse multiple branches, this increases memory usage and computational load, making it difficult to deploy on mobile devices.

Early keypoint detection was mainly used in human pose estimation. Later, with its continuous development, keypoint detection has also been applied to agricultural tasks such as fruit picking. The LS-net network[36] has been applied to strawberry picking and keypoint detection. It uses depthwise separable convolutions to improve the feature pyramid network and adopts an anchor-free decoupled head structure, which reduces computational load. However, due to its lightweight design, the model achieves low accuracy when detecting occluded and deformed strawberries. The research on pineapple fruit picking pose detection based on an improved LiteHRNet has advantages in lightweight design and high-precision pose estimation[37]. However, the introduction of modules such as LKA_Stem, D-Mixer, and MS-FFN increases the complexity of the model architecture, imposing higher requirements on deployment and maintenance, and making implementation on edge computing devices particularly challenging. TomatoPoseNet[38] adopts a keypoint-based 6D pose estimation method specifically designed for the need for non-destructive harvesting of tomatoes, achieving high keypoint detection accuracy. Its advantages lie in meeting agronomic requirements and adapting to unstructured environments, as well as effectively detecting small targets in complex environments. However, the model complexity is relatively high, posing challenges for deployment on resource-constrained devices. The multi-level feature fusion method[39] for keypoint detection of fruit tree fruiting branches adopts a top-down network architecture. Its advantages lie in maintaining a small model size and low computational requirements, making it suitable for orchard automation operations. However, its performance in handling occluded branches still needs improvement, and its robustness under extreme weather conditions also requires enhancement. In the above models, the functions are all focused on fruit localization and grasping, without introducing a ripeness judgment mechanism prior to the picking stage.

C.MULTI-TASK LEARNING IN COMPUTER VISION

The core challenge of multi-task learning lies in designing efficient parameter sharing mechanisms that fully exploit shared knowledge across tasks while mitigating negative transfer caused by task conflicts, thereby preventing overall performance degradation. Around this problem, multi-task network architectures have roughly evolved through three developmental stages: static shared fusion, global dynamic gating, and explicit structural separation with hierarchical routing. Misra *et al.* [40] proposed Cross-Stitch Networks at CVPR 2016, which perform learnable linear fusion of activations at the same layer across parallel task-specific networks, enabling mutual feature borrowing among tasks. However, the fusion weights in this model remain fixed after training, lacking the ability to dynamically adjust feature fusion strategies based on sample content. To address this limitation, Ma *et al.* [41] introduced the Multi-gate Mixture-of-Experts (MMoE) at KDD 2018, incorporating the Mixture-of-Experts structure into the multi-task learning framework: all tasks share a set of expert sub-networks while each task is equipped with an independent gating network that dynamically generates expert weight allocation strategies conditioned on the input samples, effectively alleviating negative transfer in scenarios where task correlation is weak. Nevertheless, MMoE fails to explicitly distinguish between shared representations and task-private representations, making it difficult to mitigate the typical "seesaw phenomenon" in recommendation systems, where improving one task's performance tends to degrade others. To tackle this issue, Tang *et al.* [42] proposed the Progressive Layered Extraction (PLE) model at RecSys 2020, which explicitly separates shared components and task-specific components in the network architecture and introduces a progressive routing mechanism to extract and separate deeper semantic

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knowledge in a layer-wise manner, enhancing joint representation learning and task information routing capabilities, with significant online gains demonstrated in large-scale industrial scenarios such as Tencent's video recommendation system.

III. METHODS

A. OVERALL NETWORK ARCHITECTURE

To address the multi-task collaborative challenge of cherry maturity detection and grasping keypoint localization in complex orchard environments, this paper proposes a gating-mechanism-based multi-level task collaborative enhancement network, named YOLOv8-Cherry. Built upon the YOLOv8n baseline framework, the network integrates three core components—the Dual-Task Balanced C2f module (TBC2f), the Task Harmonic Attention Module (THAM), and the Dual-Task Synergistic Attention Module (DTSAM)—thereby establishing a complete collaborative optimization pipeline that spans from the backbone network to the detection head.

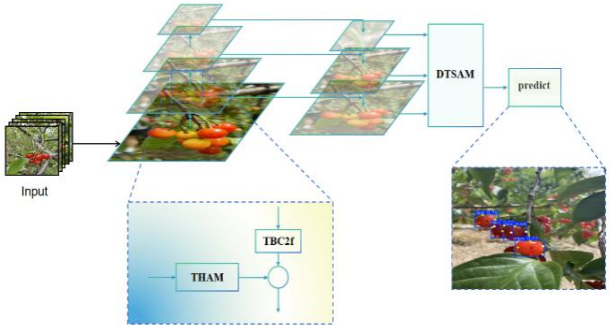


FIGURE 1. YOLOv8-Cherry structure diagram

As shown in Fig. 1, the network takes an RGB image of 640×640 resolution as input. The input is successively processed through the backbone network to extract multi-scale features, followed by task-adaptive modulation via THAM at the deep feature layers, and then fed into the neck network for bidirectional feature fusion. In the shallow layers of the neck, the TBC2f module replaces the original C2f module to achieve dynamic balancing between detection and keypoint tasks during the fine-grained feature extraction stage. Finally, after enhancement by the DTSAM module, the features at each scale are delivered to three detection heads, which output detection results and keypoint predictions in parallel. The output includes bounding-box information for two categories (ripe cherries and semi-ripe cherries), along with the normalized coordinates and visibility status of five keypoints (top, bottom, left, right, and the center of the fruit equatorial point).

The overall architecture follows a hierarchical collaborative principle that spans from deep semantic information to shallow detailed features. Specifically, THAM is inserted after the P4 (feature map scale at 1/16 of the input image) and P5 (1/32) layers in the backbone network to enhance the dual-task representational capacity of deep semantic features. In the neck network, the shallow P3 (1/8) layer is prioritized for optimization—three TBC2f modules are adopted to completely replace the original C2f modules, followed by a DTSAM module, so as to achieve dynamic balancing between detection and keypoint tasks during the fine-grained feature extraction stage. Additionally, a DTSAM module is prepended to each of the P3, P4, and P5 detection heads, forming a multi-scale task collaboration chain. This hierarchical design ensures that all feature representations, from low-level to high-level, effectively balance the dual-task requirements while preserving the efficient inference characteristics of a single-stage detector.

The forward propagation of the network can be formalized as follows. After preprocessing, the input image is fed into the backbone network, where task-adaptive attention modulation is applied via THAM at the deep P4 and P5 feature layers. Subsequently, the feature maps enter the neck network, which employs a path-aggregation structure to perform bidirectional top-down and bottom-up feature fusion, generating enhanced P3, P4, and P5 feature layers. Within the P3 layer, three TBC2f modules dynamically balance the dual-task feature representations during the detailed feature extraction stage. Afterward, the P3, P4, and P5 layers are each followed by DTSAM to further promote complementarity and synergy between the dual-task features at the detection-head level. Finally, the three detection heads produce detection and keypoint results in parallel. The internal designs of the individual modules are detailed in the subsequent subsections.

B. DUAL-TASK BALANCED C2F MODULE (TBC2F)

The Dual-Task Balanced C2f Module (TBC2f) is a deep reconstruction of the original C2f module in YOLOv8, aimed at achieving dynamic balancing between detection and keypoint tasks during the early stage of feature extraction. As shown in Fig. 2, while retaining the basic structure of the original C2f (initial convolution, channel splitting, multiple Bottleneck residual blocks, and final convolution), TBC2f introduces a parallel dual-task enhancement mechanism and a lightweight task-interaction gating network after each Bottleneck block. This enables the feature extraction process to adaptively adjust feature representations according to the different demands of the two tasks.

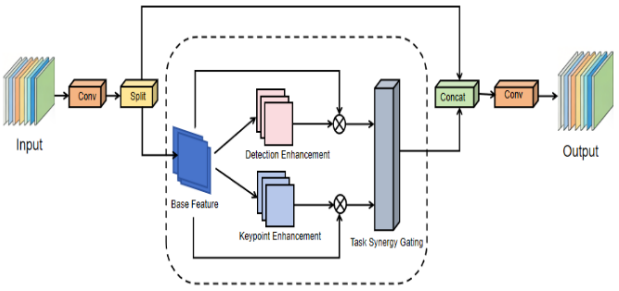


FIGURE 2. Structure of the TBC2f Module

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Base feature extraction: For the input feature X, the baseline C2f structure is first applied to extract the base feature Fbase (1). The baseline C2f adopts a cross-stage partial connection design, which splits the feature channels into two parts—one part is directly passed to the output, while the other part is processed through multiple Bottleneck blocks and then concatenated. This design not only maintains smooth gradient flow but also enables multi-scale feature fusion.

$$F_{base} = C2f_{base}(\mathrm{X}) \tag{1}$$

Parallel dual-task enhancement: On the output of each Bottleneck block, two enhancement branches are executed in parallel. The detection enhancement branch (2) adopts a multi-scale convolution fusion strategy: first, a 1×1 convolution is used to halve the number of channels to reduce computational cost; then, a standard 3×3 convolution and a dilated convolution with dilation rate 2 are applied in parallel to capture spatial information from different receptive fields. After concatenating the outputs of the two convolutions along the channel dimension, a 1×1 convolution followed by a Sigmoid activation is applied to generate the detection attention map Adet. The keypoint enhancement branch (3) adopts depthwise separable convolution combined with channel attention: first, a 3×3 depthwise separable convolution is used to extract spatial features, significantly reducing the number of parameters; then, an SE module (global average pooling, two fully connected layers, and Sigmoid) is employed to recalibrate channel-wise feature responses; finally, a 1×1 convolution and Sigmoid produce the keypoint attention map Apose.

$$\begin{gathered} F_{det} = Conv_{1\times1}(F_{base}) \\ F_{ms}^{det} = Conv_{3\times3}(F_{det}) \oplus DilatedConv_{3\times3,d=2}(F_{det}) \\ A_{det} = \sigma\left(Conv_{1\times1}(F_{ms}^{det})\right) \end{gathered} \tag{2}$$

$$\begin{gathered} F_{pose} = DWConv_{3\times3}(F_{base}) \\ F_{se}^{pose} = SE(F_{pose}) \\ A_{pose} = \sigma\left(Conv_{1\times1}(F_{se}^{pose})\right) \end{gathered} \tag{3}$$

Lightweight Task-Interaction Gating. To dynamically regulate the information flow between the two tasks, TBC2f concatenates the detection-enhanced features and the keypoint-enhanced features along the channel dimension. This concatenated feature map is first compressed spatially via global average pooling (GAP), yielding a C-dimensional vector. Two fully connected (FC) layers are then applied to learn the gating weights: the first FC layer reduces the channel dimension by a factor of four with ReLU activation, and the second FC layer outputs two channels followed by Sigmoid activation. These produce the gating weights Gdet and Gpose (4).

$$\begin{aligned} &[G_{det}, G_{pose}] \\ &= \sigma\left(Conv_{1\times1}\left(ReLU\left(Conv_{1\times1}\left(GAP([A_{det}, A_{pose}])\right)\right)\right)\right) \end{aligned} \tag{4}$$

Remark. In practical implementations, the two FC layers are often equivalently realized as 1×1 convolutional layers, as the GAP output is a 1×1×C tensor; mathematically, however, the FC formulation is the precise one.

Dynamic fusion: Based on the gating weights, the task-interaction gating network simultaneously produces three unnormalized logits λ_{det}, λ_{pose}, and λ_{int}, which respectively indicate the contributions of the detection branch, the keypoint branch, and their interaction branch. The final enhanced feature is then computed as the weighted summation of three components: the gated detection feature, the gated keypoint feature, and the direct interaction feature (element-wise addition of the two enhanced features). The corresponding task weights α, β, and γ are derived by applying the Softmax function (5) to the three logits. For the TBC2f module containing multiple Bottleneck blocks, each block sequentially performs the above operations; the branch features from all blocks are eventually concatenated and passed through the output convolution layer to adjust the channel number. where $\odot$ denotes element-wise multiplication, $\oplus$ denotes element-wise addition, and the sum of the three Softmax-normalized weights satisfies α + β + γ = 1.

$$\begin{gathered} \alpha = \frac{e^{\lambda_{det}}}{e^{\lambda_{det}} + e^{\lambda_{pose}} + e^{\lambda_{int}}}, \beta = \frac{e^{\lambda_{pose}}}{e^{\lambda_{det}} + e^{\lambda_{pose}} + e^{\lambda_{int}}}, \gamma = \frac{e^{\lambda_{int}}}{e^{\lambda_{det}} + e^{\lambda_{pose}} + e^{\lambda_{int}}} \\ F_{enhanced} = \alpha \cdot (G_{det} \odot F_{det}) + \beta \cdot \left(G_{pose} \odot F_{pose}\right) + \gamma \cdot \left(F_{det} \oplus F_{pose}\right) \end{gathered} \tag{5}$$

C. TASK HARMONIC ATTENTION MODULE (THAM)

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The Task Harmonic Attention Module (THAM) is an attention mechanism specifically designed for multi-task learning, aiming to coordinate the attention distributions of detection and keypoint tasks at the deep feature level. As shown in Fig. 3, THAM first projects the input features into a low-dimensional space via a shared feature extractor, then executes two task-specific attention generation branches in parallel, and finally adaptively fuses the attention maps from the two branches through a harmonic weight generation gating mechanism.

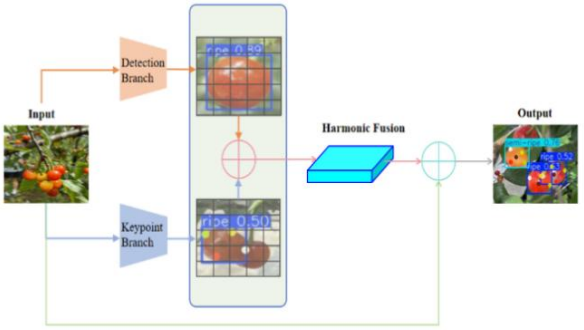


FIGURE 3. Structure of the Task Harmonic Attention Module (THAM)

Shared feature extraction: A 1×1 convolution (with batch normalization and ReLU) is used to reduce the input channel number to one eighth of the original (with a minimum of 8 channels), preserving sufficient information for attention generation while reducing the subsequent computational cost (6).

$$F_{shared} = ReLU\left(BN\left(Conv_{1\times1}(X)\right)\right) \tag{6}$$

Dual-task attention generation (7). The detection attention branch adopts a standard 3×3 convolution to extract spatial contextual information and generate the detection attention map. This branch uses a 1×1 convolution to restore the channel number to the input channel size, followed by a Sigmoid function to map the values to the range of 0 to 1. The keypoint attention branch adopts depthwise separable convolution with channel expansion operations to enhance fine-grained perceptual capability. Specifically, a 3×3 depthwise convolution is first used to extract spatial features (with the parameter count reduced to 1/C of that of a standard convolution), and then two consecutive 1×1 convolutions are applied to perform channel expansion and compression (first doubling the channel number, then restoring to the original value), finally followed by a Sigmoid function to produce the keypoint attention map.

$$\begin{aligned} A_{det} &= \sigma\left(Conv_{1\times1}\left(ReLU\left(BN\left(Conv_{3\times3}(F_{shared})\right)\right)\right)\right) \\ F'_{pose} &= DWConv_{3\times3}(F_{shared}) \\ F_{exp}^{pose} &= Conv_{1\times1}\left(ReLU\left(BN\left(Conv_{1\times1}(F'_{pose})\right)\right)\right) \\ A_{pose} &= \sigma\left(Conv_{1\times1}\left(F_{exp}^{pose}\right)\right) \end{aligned} \tag{7}$$

Harmonic weight generation gating: The core innovation of THAM lies in dynamically predicting three harmonic weights based on the global statistical information of the input features, which respectively represent the contribution degrees of detection-exclusive, keypoint-exclusive, and dual-task shared components. The harmonic weight generator first compresses the spatial features of each channel into a single average value via global average pooling, forming a C-dimensional feature vector. It then learns weight prediction through two fully connected layers: the first fully connected layer reduces the channel number to one quarter of the original (with a minimum of 8), using ReLU activation; the second fully connected layer outputs three channels corresponding to the three harmonic weights. Finally, the three output values are normalized via the Softmax function to ensure that they sum to 1. These three gating weights are used to fuse the two task-specific attention maps into a harmonic attention map through weighted summation, where the attention for the dual-task shared component is taken as the average of the two attention maps.

Feature enhancement and residual fusion: The harmonic attention map is combined with the input features to generate enhanced features, computed as the input features multiplied by (1 plus the weighted harmonic attention map). The final output is fused with the original features via a learnable gating parameter for residual fusion. The residual connection ensures that even if the attention mechanism fails, the original features can still be completely passed through, avoiding performance degradation.

The harmonic gating mechanism of THAM enables the network to adaptively adjust its task attention strategy according to the input content: when the cherry targets in the image are clear and occlusions are minor, the dual-task shared weight is relatively high, and the two tasks share the attention resources; when the targets are blurred or heavily occluded, the network can increase the detection-exclusive or keypoint-exclusive weight to concentrate resources on the more feasible task. This adaptive capability significantly improves the model's robustness in complex orchard environments.

D. DUAL-TASK SYNERGISTIC ATTENTION MODULE (DTSAM)

The Dual-Task Synergistic Attention Module (DTSAM) achieves task synergy at the detection-head level, mining complementary information between detection and keypoint tasks through an explicit collaborative learning gating mechanism. As shown in Fig. 4, DTSAM consists of four core components: the base attention layer, task-specific enhancers, collaborative gating, and a collaborative learner, with all components organically coordinated through gating mechanisms.

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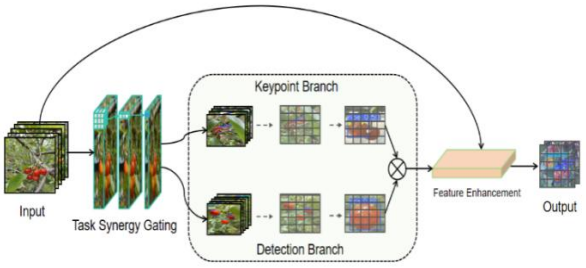


FIGURE 4. Structure of the Dual-Task Synergistic Attention Module (DTSAM)

Base attention layer: Provides a shared attention prior for the two tasks. It employs two consecutive 1×1 convolutions: the first reduces the channel number to one eighth of the original (with a minimum of 8), using batch normalization and ReLU activation; the second restores the channel number to the original value, and finally a Sigmoid function is applied to generate the base attention map A_{base}.

Task-specific enhancers are optimized separately for the two tasks. Detection spatial perceiver: A 3×3 dilated convolution with dilation rate 2 is used to expand the receptive field to 5×5 while keeping the parameter count unchanged, enabling the capture of broader contextual information, which is particularly important for determining object boundaries. The dilated convolution is followed by batch normalization, ReLU activation, a 1×1 convolution, and a Sigmoid function to generate the detection spatial attention map A_{detsp}.

Keypoint detail perceiver: A 3×3 depthwise separable convolution is first used to extract local features, followed by two 1×1 convolutions for channel expansion and compression (first doubling the channel number, then restoring to the original value), with batch normalization and ReLU activation applied in between, and finally a Sigmoid function is applied to generate the keypoint detail attention map A_{posedt}. This design enables the capture of detail features at different granularities, which is crucial for precise keypoint localization.

Collaborative gating: A lightweight network generates two-channel gating weights to dynamically adjust the participation degrees of the two tasks. As shown in (8), collaborative gating first obtains the global statistical information of the input features via global average pooling, and then learns the gating weight prediction through two fully connected layers: the first fully connected layer reduces the channel number to one eighth of the original (with a minimum of 4), using ReLU activation; the second fully connected layer outputs two channels, which are mapped to the range of 0 to 1 by a Sigmoid function, corresponding to the gating weights G_{det} and G_{pose} for the detection task and the keypoint task, respectively.

Gated enhanced features: The detection enhanced feature $F_{detgated}$ and the keypoint enhanced feature $F_{posegated}$ are respectively computed as the input features multiplied by the average of the base attention map and the gated task-specific attention map (8).

$$[G_{det}, G_{pose}] = \sigma\left(Conv_{1\times1}\left(ReLU\left(Conv_{1\times1}(GAP(X))\right)\right)\right)$$
$$F_{gated}^{det} = X \odot \left(\frac{A_{base} + G_{det} \cdot A_{sp}^{det}}{2}\right)$$
$$F_{gated}^{pose} = X \odot \left(\frac{A_{base} + G_{pose} \cdot A_{dt}^{pose}}{2}\right) \quad (8)$$

Collaborative learner: Taking the concatenated dual-task features as input, it learns a synergistic attention map between the two tasks. First, the detection enhanced feature Fdetgated and the keypoint enhanced feature Fposegated are concatenated along the channel dimension, doubling the number of channels. Then two 1×1 convolutions are applied to learn the synergistic patterns: the first reduces the channel number to one quarter of the original with ReLU activation, and the second restores the channel number to the original value, followed by a Sigmoid function to generate the synergistic attention map Asyenergy, explicitly modeling the complementary relationships between the two tasks.

Output feature fusion: As shown in (9), the output feature is obtained by adding three weighted components to the original feature: the detection enhanced feature, the keypoint enhanced feature, and the synergistically enhanced feature (computed as the element-wise product of the synergistic attention map with the element-wise sum of the detection enhanced feature and the keypoint enhanced feature). The learnable weights w_d, w_k, w_s are generated from the learnable parameter λ via a Sigmoid function. This fusion scheme ensures that even if the feature response of one task is relatively weak, it can be strengthened through the synergistic mechanism while preserving the integrity of the original features.

$$A_{synergy} = \sigma\left(Conv_{1\times1}\left(ReLU\left(Conv_{1\times1}\left(\left[F_{gated}^{det}, F_{gated}^{pose}\right]\right)\right)\right)\right)$$
$$w_d = \sigma(\lambda_d), w_k = \sigma(\lambda_k), w_s = \sigma(\lambda_s)$$
$$F_{out} = X + w_d \cdot F_{gated}^{det} + w_k \cdot F_{gated}^{pose} + w_s \cdot \left(\left(F_{gated}^{det} \oplus F_{gated}^{pose}\right) \odot A_{synergy}\right) \quad (9)$$

The collaborative gating mechanism of DTSAM is capable of identifying complementary regions between the two task features. When the detection features exhibit strong responses in a certain region while the keypoint features respond weakly, the synergistic attention map assigns higher weights to the keypoint features in that region, facilitating knowledge transfer between the tasks. This design significantly improves keypoint localization accuracy under small objects, ambiguous boundaries, and complex backgrounds.

IV. EXPERIMENTS

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A. DATASET

1) DATASET COLLECTION

To address the practical problem of cherry harvesting, we collected 1,100 images of crystal cherries on April 8, 2025, at the open-air cherry cultivation base in Huangjiashan Village, Bagou Town, Qianwei County, Leshan City, Sichuan Province, China. On April 16, 2025, we captured 1,000 images of red agate cherries in Hanyuan County, Ya'an City, Sichuan Province, China. The original dataset comprised 2,100 images. All images were taken with a handheld camera at a shooting distance of approximately 200 mm to 1,000 mm. The collected images cover complex scenarios including cherries under different lighting conditions, branches and leaves occluding cherries, cherries occluding each other, and cherries at different maturity stages, as illustrated in Fig. 5. The image resolution is 640 × 640 pixels, and the images are stored in JPG format.

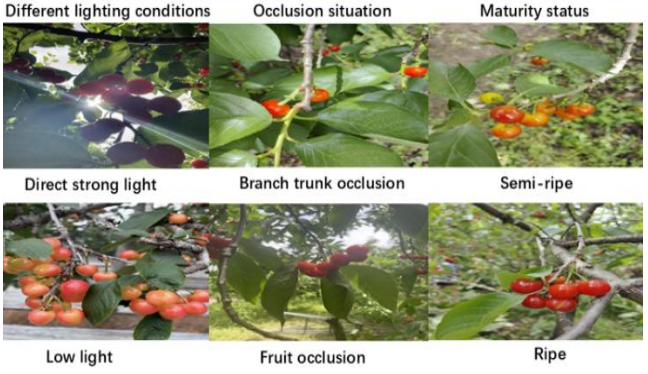


FIGURE 5. **Typical photos of the cherry dataset are presented in the figure.**

2) IMAGE ANNOTATION AND DATASET CONSTRUCTION

This study used Roboflow, a web-based dataset annotation tool, to label cherry regions and five keypoints. The data annotation is shown in Fig. 6. We took the cherry as the detection object and annotated it with bounding boxes. For maturity classification, we define two categories based on the surface color of cherries: 'semi-ripe' (encompassing both green and pinkish-red cherries with partial or no coloration) and 'ripe' (deep red to dark burgundy, fully colored, suitable for harvesting). Based on the cherry's appearance structure, we selected five easily identifiable parts—"calyx," "stem_end," "equator_1," "equator_2," and "equator_3"—as keypoints. To better reflect the actual cherry picking scenario, we did not annotate cherries that were too far away with bounding boxes. When annotating keypoints, keypoints that were invisible due to mutual occlusion were marked as "1", keypoints that were invisible due to truncation were marked as "0", and other keypoints were marked as "2" according to the annotation standards. It is important to note that, in order to locate multiple objects and their corresponding keypoints within the same image, the annotation process must first annotate the bounding boxes for the cherry regions, followed by the annotation of the five keypoints. The annotation tool can automatically export the data as a TXT file, where each line represents an object and its keypoints within the image, comprising a total of 20 columns. The first column is the category information. In this study, semi-ripe targets are labeled as "semi-ripe", and ripe targets are labeled as "ripe". Columns 2 to 5 represent the coordinates of the bounding box (Xc, Yc) as well as the width and height of the bounding box (Wb, Hb). Columns 6 to 20 contain the normalized coordinates of the five keypoints and the keypoint status (Xki, Yki, a). All images were annotated by our research team. To ensure labeling consistency, each image was carefully reviewed by at least two team members, and any disagreements were resolved through discussion. For borderline cases where color discrimination was ambiguous, a senior agricultural expert was consulted for the final decision. To improve the detection robustness of the model in complex environments, we performed data augmentation on the dataset. The augmentation methods were applied exclusively to the training set after the dataset split to prevent data leakage. The augmentation methods included auto-orientation, unequal stretching, color adjustment, brightness variation, blurring, and random noise addition. After data augmentation, the training set contained 3,942 images (2,760 for training, 789 for validation, and 393 for testing). The training set contains approximately balanced samples of semi-ripe and ripe cherries, with similar class distributions maintained across the validation and test splits. The overall mAP@0.5 reported in the experimental section is computed over test samples containing both semi-ripe and ripe cherries, thereby reflecting the model's maturity detection capability across the full ripeness spectrum encountered in real harvesting scenarios.

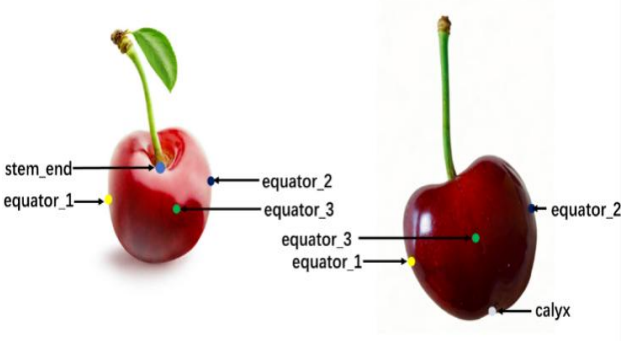


FIGURE 6. **Keypoint annotation diagram.**

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B. EXPERIMENTAL ENVIRONMENT

All experiments in this study were conducted on a computer equipped with the Windows 11 operating system. The hardware configuration was as follows: the processor was a 13th Gen Intel Core i7-13650HX with a base frequency of 2.60 GHz; the memory capacity was 24 GB; and the graphics processing unit was an NVIDIA GeForce RTX 4060 with 8 GB of VRAM. The software environment was based on Python 3.8.20. Model training was optimized using the stochastic gradient descent algorithm, with key hyperparameters set as follows: the number of training epochs was 300, the batch size was 36, the number of data loading workers was 1, and the initial learning rate was 0.005. The input image size was uniformly resized to 640×640 pixels. Evaluation metrics included precision, recall, mAP@0.5, and mAP@0.5:0.95 for bounding boxes, as well as precision, recall, mAP@0.5, and mAP@0.5:0.95 for keypoints. Additionally, the model's floating-point operations (FLOPs), parameter count, and inference speed were also calculated.

C. EVALUATION METRICS

To comprehensively evaluate the model performance, we adopt Precision, Recall, mAP@0.5, mAP@0.5:0.95, FPS, model size, and GFLOPs as evaluation metrics. Precision is the ratio of samples predicted as positive that are actually positive. Recall is the ratio of samples predicted as positive among all actual positive samples. Here, TP (True Positive) denotes that the model predicts a positive class and the actual class is also positive. FP (False Positive) denotes that the model predicts a positive class but the actual class is negative. FN (False Negative) denotes that the model predicts a negative class but the actual class is positive. mAP@0.5 is the mean Average Precision over all classes when the IoU threshold is set to 0.5, as shown in (10):

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$AP = \int_0^1 Precision(Recall)dRecall$$

$$mAP = \frac{\sum_{i=1}^{N} AP_i}{N} \tag{10}$$

D. COMPARATIVE EXPERIMENT

1) COMPARISON WITH THE YOLO SERIES

To comprehensively evaluate the performance of YOLO-Pose (denoted as Ours in the table), we conducted thorough comparisons with models in the YOLO series. In the experiments, the Cherry dataset was used, and all models were trained for 300 epochs to ensure sufficient convergence and stable performance. The detailed experimental results are presented in Table 1.

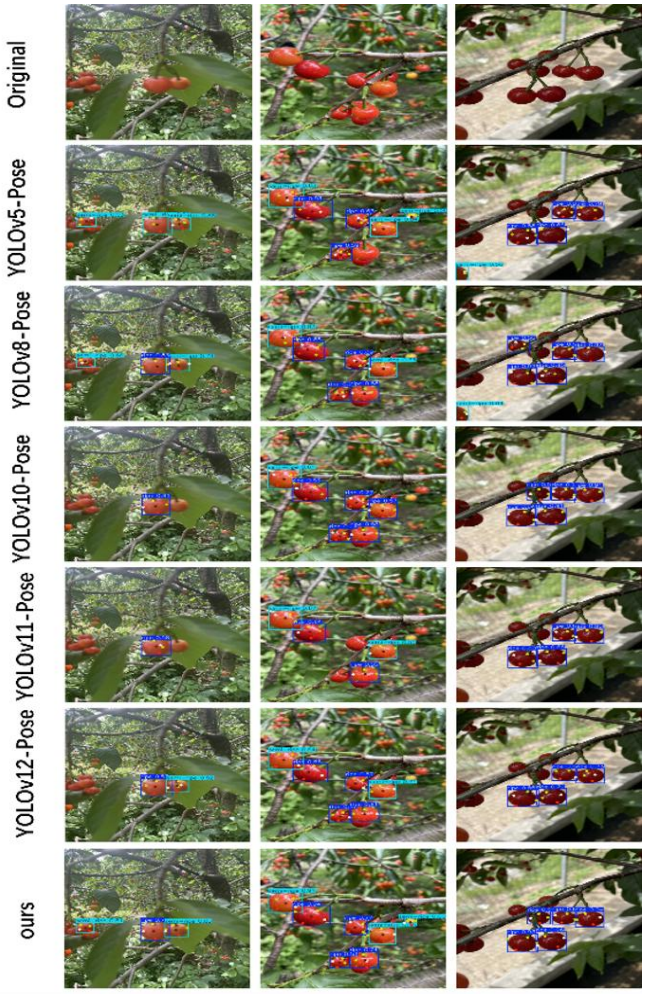


FIGURE 7. Visualization results of comparative experiments

The experimental results are presented in Table 1. Our model achieves a Box mAP@0.5 of 0.803 on the Cherry dataset, which is higher than those of YOLOv5-Pose, YOLOv8-Pose, YOLOv10-Pose, YOLOv11-Pose, and YOLOv12-Pose by 4.6%, 6.6%, 3.0%, 3.8%, and 4.1%, respectively. Our model attains a Box Precision of 0.704, outperforming the corresponding values of YOLOv5-Pose, YOLOv8-Pose, YOLOv10-Pose, YOLOv11-Pose, and YOLOv12-Pose by 6.2%, 5.5%, 0.6%, 1.0%, and 5.0%, respectively. The Box Recall of our model on the Cherry dataset is 0.776, which is higher than those of YOLOv5-Pose, YOLOv10-Pose, YOLOv11-Pose, and YOLOv12-Pose by 1.7%, 7.5%, 9.1%, and 4.1%, respectively; however, it is 0.5% lower than that of YOLOv8-Pose. For the Pose metrics, our model obtains a Pose mAP@0.5 of 0.773, exceeding those of YOLOv5-Pose, YOLOv8-Pose, YOLOv11-Pose, and YOLOv12-Pose by 6.6%, 6.9%, 2.8%, and 5.6%, respectively. The Pose Precision of our model is 0.689, which is higher than those of YOLOv5-Pose, YOLOv8-Pose, and YOLOv12-Pose by 5.6%, 6.0%, and 5.1%, respectively. In terms of model size, our model has more Parameters and a larger Model Size than YOLOv5-Pose, YOLOv10-Pose, YOLOv11-Pose, and YOLOv12-Pose. Our model achieves a higher FPS than

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YOLOv11-Pose and YOLOv12-Pose, and although it is lower than those of YOLOv5-Pose, YOLOv8-Pose, and YOLOv10-Pose, it still meets the real-time requirements for cherry detection. Our model has GFLOPs of 7.6, which is slightly higher than those of YOLOv5-Pose, YOLOv10-Pose, YOLOv11-Pose, and YOLOv12-Pose; nevertheless, our model still demonstrates superior overall performance. The consistent accuracy improvements across both detection and keypoint localization tasks can be attributed to the proposed multi-level dual-task collaborative mechanism. Existing YOLO-Pose models simply concatenate detection and pose regression branches without explicit task interaction, which leads to feature competition and task conflicts under complex orchard conditions such as illumination variation, leaf occlusion, and fruit overlapping. In contrast, the TBC2f module dynamically balances the two tasks during feature fusion, the THAM module adaptively enhances task-specific shallow semantic features, and the DTSAM module exploits cross-task complementary information at the detection head. This hierarchical collaborative optimization fundamentally alleviates the inherent seesaw problem between detection and keypoint localization, enabling simultaneous performance gains in both tasks rather than improving one at the expense of the other. The slight 0.5% recall gap compared with YOLOv8-Pose and the moderate increases in parameters, model size, and GFLOPs are reasonable trade-offs. The embedded collaborative modules introduce additional task-interaction modeling overhead, but this cost is justified by the substantial and consistent accuracy gains across all primary metrics (Box mAP@0.5, Box Precision, Pose mAP@0.5, and Pose Precision). Moreover, the model maintains a competitive inference speed and fully satisfies the real-time perception requirements of agricultural picking robots operating in field environments. In summary, the quantitative results demonstrate that YOLOv8-Cherry achieves a favorable balance among detection accuracy, keypoint localization precision, processing speed, and model complexity. The proposed task collaborative optimization strategy provides an effective solution for robust dual-task perception in complex orchard environments and offers practical reference value for agricultural robot visual sensing systems. Visualization results are shown in Fig. 7, which further intuitively confirms the superior anti-interference capability and localization stability of the proposed method.

TABLE 1. Comparative experiments of different models on the Cherry dataset.

Model	Detection				Keypoint				Parameters	Model Size(MB)	FPS	GFLOPs
	mAP@0.5	mAP@0.5-0.95	Precision	Recall	mAP@0.5	mAP@0.5-0.95	Precision	Recall				
YOLOv5-Pose	0.757	0.603	0.642	0.759	0.707	0.502	0.633	0.723	2,581,235	5.23	58.8	7.4
YOLOv8-Pose	0.737	0.617	0.649	0.781	0.704	0.369	0.629	0.756	3,078,323	6.16	46.5	8.3
YOLOv10-Pose	0.773	0.635	0.698	0.701	0.770	0.587	0.698	0.701	2,337,843	4.82	46.3	6.8
YOLOv11-Pose	0.765	0.610	0.694	0.685	0.745	0.579	0.689	0.680	2,654,827	5.42	29.8	6.6
YOLOv12-Pose	0.762	0.623	0.654	0.735	0.717	0.432	0.638	0.716	2,629,403	5.46	29.9	6.6
Ours	0.803	0.665	0.704	0.776	0.773	0.563	0.689	0.759	2,991,631	6.00	43.1	7.6

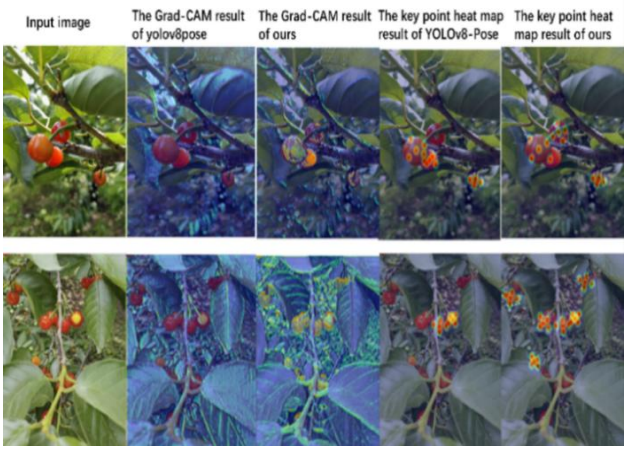


FIGURE 8. Comparison results between Ours and YOLOv8-Pose

As shown in Fig. 8, in the Grad-CAM heatmap comparison, the activation regions of our model are more concentrated and completely cover the cherry fruits, with significantly reduced background interference, indicating a stronger feature focusing capability for cherry targets. In the keypoint heatmap comparison, the response regions of our model are clearer, highly consistent with the actual keypoint locations of the cherries, and also exhibit higher response intensity, demonstrating superior accuracy and reliability in keypoint localization. Overall, by introducing the TBC2f, THAM, and DTSAM modules, our model significantly enhances feature representation and localization capability for cherry targets, providing more precise visual support for cherry picking tasks.

2) COMPARISON WITH HDPNET, RT-DETR, AND YOLOV8

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To better evaluate the detection capability of our model for cherry ripeness, we compare our model with HDPNet, RT-DETR, and YOLOv8. The experiments are conducted on the Cherry dataset with a total of 300 training epochs. The experimental results are shown in Table 2.

TABLE 2. Comparative experimental results of different models for cherry ripeness detection.

Model	mAP@0.5	mAP@0.5-0.95	Precision	Recall	Parameters	Model Size(MB)	FPS	GFLOPs
HDPNet	0.755	0.608	0.712	0.672	2,564,147	8.30	46.9	9.0
RT-DETR	0.655	0.556	0.634	0.711	32,810,186	66.20	22.1	103.4
YOLOv8	0.737	0.607	0.650	0.715	3,011,238	5.99	81.9	8.2
Ours-Detect	0.788	0.645	0.734	0.716	2,490,161	5.01	69.4	7.1
Ours	0.803	0.665	0.704	0.776	2,991,631	6.00	43.1	7.6

The experimental results are shown in Table 2. The Box mAP@0.5 achieved by Ours-Detect is 0.788, which is 3.3%, 13.3%, and 5.1% higher than those of HDPNet, RT-DETR, and YOLOv8, respectively. The Box Precision of Ours-Detect is 2.2%, 10.0%, and 8.4% higher than those of HDPNet, RT-DETR, and YOLOv8, respectively. The Recall of our model is 0.716, which is 4.4%, 0.5%, and 0.1% higher than those of HDPNet, RT-DETR, and YOLOv8, respectively. The Parameters of our model are 2,490,161, which is lower than those of HDPNet, RT-DETR, and YOLOv8. The Model Size of our model is 5.01 MB, which is smaller than those of HDPNet, RT-DETR, and YOLOv8. The FPS of our model is 69.4. Although lower than that of YOLOv8, our model still meets the real-time requirements for cherry detection. The GFLOPs of our model are also lower than those of HDPNet, RT-DETR, and YOLOv8. In summary, our model achieves a good balance among detection accuracy, real-time performance, and lightweight design for cherry ripeness detection. After incorporating keypoint detection into Ours-Detect, the Box mAP@0.5 reaches 0.803, which is 1.5% higher than that of the original Ours-Detect. This demonstrates that the introduction of keypoint detection improves the detection accuracy of our model. Therefore, the fusion of the keypoint detection branch with the object detection branch enhances the overall detection performance of the model.

3) COMPARATIVE EXPERIMENTS WITH THE YOLO FAMILY ON THE CHERRY TOMATO DATASETS.

We also compare our model with the YOLO series on the Cherry Tomato Datasets[43]. Since only the training and validation sets are publicly available for this dataset, we split the dataset into training, validation, and test sets with a ratio of 7:2:1 to ensure experimental fairness. The experiments are conducted for 300 epochs, and the results are shown in Table 3.

The experimental results are shown in Table 3. On the Cherry Tomato Datasets, our model achieves a Box mAP@0.5 of 0.833, which is 7.3%, 1.2%, and 3.7% higher than those of YOLOv8-Pose, YOLOv11-Pose, and YOLOv12-Pose, respectively. The Pose mAP@0.5 of our model is 0.810, outperforming YOLOv8-Pose, YOLOv11-Pose, and YOLOv12-Pose by 5.9%, 4.6%, and 5.9%, respectively. The FPS of our model is higher than those of YOLOv8-Pose and YOLOv12-Pose, and although it is lower than that of YOLOv11-Pose, it still meets the real-time requirements for fruit detection. This experiment demonstrates that our model possesses good generalization capability.

TABLE 3. Comparison experiments of different models on the Cherry Tomato Datasets.

Model	Detection				Keypoint				Parameters	Model Size (MB)	FPS	GFLOPs
	mAP@0.5	mAP@0.5-0.95	Precision	Recall	mAP@0.5	mAP@0.5-0.95	Precision	Recall				
YOLOv8-Pose	0.760	0.419	0.872	0.627	0.751	0.745	0.821	0.665	3,078,323	6.16	95.2	8.3
YOLOv11-Pose	0.821	0.437	0.820	0.739	0.764	0.756	0.818	0.697	2,654,827	5.42	169.5	6.6
YOLOv12-Pose	0.796	0.428	0.863	0.672	0.751	0.745	0.814	0.680	2,629,403	5.46	65.8	6.6
Ours	0.833	0.471	0.861	0.751	0.810	0.801	0.851	0.731	2,991,631	6.00	136.9	7.6

4) COMPARISON WITH OTHER MAINSTREAM KEYPOINT DETECTION MODELS

We also compared our model with several mainstream models in the field of keypoint detection, including ViTPose[44], RTMPose[45], and Lite-HRNet-30[46].In the comparative experiments, the input image size was uniformly set to 640 × 640 pixels. To ensure a fair comparison of keypoint localization performance, all methods adopted ground-truth bounding boxes as the input for keypoint detection. The evaluation metric was PCK@0.15, where a prediction was

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considered correct if the Euclidean distance between the predicted keypoint and the ground-truth keypoint was less than 0.15 times the cherry equatorial diameter. For our model, the detection branch was removed and only the keypoint localization branch was retained, which also took ground-truth bounding boxes as input. This eliminated the cumulative error introduced by the detection stage, enabling a fair comparison of keypoint estimation accuracy across all models. The experimental results are presented in Table 4.

TABLE 4. Comparison experiments of different models on the Cherry Tomato Datasets.

Model	PCK@0.15	FPS	Parameters
ViTPose	0.817	22.9	85,822,165
RTMPose	0.770	33.2	5,814,941
Lite-HRNet-30	0.755	15.2	1,763,763
Ours	0.841	32.3	2,991,631

The experimental results are shown in Table 4. RTMPose achieves the fastest inference speed, and there remains a certain gap between our model and it. However, in terms of accuracy, our model achieves a PCK@0.15 of 0.841, which is 2.4%, 7.1%, and 8.6% higher than those of ViTPose, RTMPose, and Lite-HRNet-30, respectively. Meanwhile, our model has only 2.99M parameters, which is only slightly higher than that of Lite-HRNet-30. These results indicate that our end-to-end model achieves the best balance among the three dimensions of accuracy, real-time performance, and lightweight design, making it particularly suitable for deployment scenarios with resource constraints or high throughput requirements.

E. ABLATION EXPERIMENT

To validate the effectiveness of YOLOv8-Cherry, we conducted ablation studies on the Cherry dataset. We used YOLOv8-Pose as the baseline model. Under the same experimental settings, we gradually added our proposed TBC2f, THAM, and DTSAM modules and conducted comprehensive comparisons to analyze the changes in model performance before and after each improvement. In the experiments, the number of training epochs was set to 300. As shown in Table 5, the symbol ✓ indicates that the corresponding module was added to the model.

As shown in Table 5, after replacing the C2f modules in the P4 and P5 layers of the backbone network of the YOLOv8-Pose model with the THAM module, the visualization results are presented in Figure 9. For the detection task, mAP@0.5 and mAP@0.5-0.95 improved by 1.7% and 1.4%, respectively; Detection Precision increased by 2.2%, and keypoint Precision increased by 3.9%. However, Recall decreased slightly in both tasks. This is because the THAM module, through its task-adaptive attention mechanism, dynamically coordinates the feature focus between the detection and keypoint tasks, enhancing the representation capability of the overall target structure in deep semantic information, thereby improving the model's recognition accuracy in complex orchard environments.

After completely replacing the original C2f module in the shallow P3 layer of the prediction network of the YOLOv8-Pose model with the TBC2f module, the visualization results are shown in Figure 10. The model's detection mAP@0.5 increases from 0.737 to 0.764, an improvement of 2.7%; the keypoint mAP@0.5 increases from 0.704 to 0.74, an improvement of 3.6%. At the same time, the FPS shows a certain increase, and the number of parameters decreases from 3,078,323 to 2,485,812. This is because the TBC2f module introduces a dual-task balancing mechanism during shallow feature extraction. By employing parallel detection and keypoint enhancement branches along with a lightweight gating network, it optimizes feature representation, thereby improving the model's operational efficiency and accuracy.

After inserting the DTSAM module into the P3, Head-P4, and Head-P5 layers of the YOLOv8-Pose model, the visualization results are shown in Figure 11. For the detection task, mAP@0.5 improves from 0.737 to 0.741, an increase of 0.4%; for the keypoint task, mAP@0.5 improves from 0.704 to 0.71, an increase of 0.7%; and keypoint mAP@0.5-0.95 significantly increases from 0.369 to 0.552, a substantial improvement of 18.3%. Detection Precision increases by 2.5%, and keypoint Precision increases by 3.3%. At the same time, the FPS decreases slightly, the number of parameters increases modestly, and both detection Recall and keypoint Recall show a decline. This is because the DTSAM module performs cross-scale task collaboration optimization in the prediction network, enhancing the interaction among multi-scale features through dynamic weight allocation, which is particularly beneficial for small-target detection and precise keypoint localization.

After integrating both THAM and TBC2f into the YOLOv8-Pose model, the detection mAP@0.5 improves from 0.737 to 0.782, an increase of 4.5%; the keypoint mAP@0.5 improves from 0.649 to 0.763, an increase of 11.4%; both the model size and the number of parameters decrease substantially. Detection Precision increases by 3.3% and keypoint Precision by 4.1%, while Recall shows a slight decrease. This is because THAM coordinates semantic attention at the deep level, whereas TBC2f balances task features at the shallow level. Together, they form a hierarchical optimization chain from deep to shallow layers, yielding significant synergistic gains and greatly improving the model's accuracy and efficiency when recognizing challenging targets.

After simultaneously incorporating both the THAM and DTSAM modules, the detection mAP@0.5 improves from 0.737 to 0.763, an increase of 2.6%; the keypoint mAP@0.5 improves from 0.649 to 0.756, an increase of 10.7%; and the keypoint mAP@0.5-0.95 significantly increases from 0.369 to 0.603, an improvement of 23.4%. Detection Precision increases by 8.4%, and keypoint Precision increases by 10.1%. However, both detection Recall and keypoint Recall decrease, and FPS also shows a decline. This is because the combination

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of deep-level task-adaptive attention and the multi-scale task collaboration mechanism jointly enhances the ability to uniformly model and align feature representations of target structure and keypoint details in complex scenes, greatly assisting the model in handling various challenging environments.

After simultaneously incorporating both the TBC2f and DTSAM modules, the detection mAP@0.5 improves by 4.8%; the keypoint mAP@0.5 improves by 5.3%; and the keypoint mAP@0.5-0.95 improves by approximately 16.1%. Moreover, the FPS increases to 42.2. However, detection Recall decreases by 1.5%, and keypoint Recall decreases by 0.4%. This is because the TBC2f module performs task-balanced optimization on detailed features during the early stage of feature extraction, while the DTSAM module achieves multi-scale task collaboration in the prediction network, dynamically fusing and optimizing feature interactions across different tasks. Together, these two modules significantly improve keypoint estimation accuracy without relying on deep-level attention modules.

After integrating all three modules—THAM, TBC2f, and DTSAM—to form the complete model, the ablation results are shown in Figure 12. Compared with the original YOLOv8-Pose model, detection mAP@0.5 improves by 6.6%, keypoint Recall increases by 7.1%, detection mAP@0.5-0.95 improves by 4.8%, and keypoint mAP@0.5-0.95 increases by 22.1%. In addition, the number of parameters decreases from 3,078,323 to 2,991,631, and the model size is reduced from 6.16 MB to 6.00 MB. Although detection Recall shows a slight decline, the overall performance is substantially stronger than that of the YOLOv8-Pose model. This improvement is attributed to the complete synergistic design of the modules, which achieves efficient model compression and acceleration with almost no loss of core performance through structural sharing, intelligent allocation of attention mechanisms, and optimization of feature pathways, thereby attaining the best balance between accuracy and efficiency.

In summary, the ablation study above demonstrates that each of the three modules individually contributes positively to the overall performance. However, a critical observation is that the sum of individual mAP@0.5 improvements by THAM (+1.7%), TBC2f (+2.7%), and DTSAM (+0.4%) amounts to only 4.8%, whereas their combined integration yields a total improvement of 6.6%. This 1.8% performance surplus explicitly confirms the existence of a synergistic effect among the three modules, indicating that the performance gain primarily stems from their combined cooperative architecture rather than from any single component alone. Specifically, THAM coordinates deep semantic attention, TBC2f balances shallow task-specific feature representations, and DTSAM refines multi-scale cross-task interactions. Together, they form a hierarchical optimization chain that jointly enhances both detection and keypoint localization—a collaborative paradigm that prior multi-task YOLO designs have not explored.

TABLE 5. Results of the ablation experiment

THAM	TBC2f	DTSAM	Detection				Keypoint				Model Size (MB)	Parameters	FPS	GFLOPs
			mAP @0.5	mAP @0.5 -0.95	Precision	Recall	mAP @0.5	mAP @0.5 -0.95	Precision	Recall				
			0.737	0.617	0.649	0.781	0.704	0.369	0.629	0.756	6.16	3,078,323	46.5	8.3
✓			0.754	0.631	0.671	0.743	0.742	0.595	0.668	0.740	5.01	2,466,816	30.5	7.4
	✓		0.764	0.617	0.721	0.725	0.740	0.459	0.710	0.714	5.11	2,485,812	41.5	7.4
		✓	0.741	0.618	0.674	0.735	0.710	0.552	0.662	0.723	6.37	3,156,706	39.2	8.6
✓	✓		0.782	0.654	0.682	0.742	0.763	0.548	0.670	0.725	4.99	2,452,229	41.8	7.1
✓		✓	0.763	0.633	0.733	0.690	0.756	0.603	0.730	0.687	5.21	2,545,199	39.4	7.6
	✓	✓	0.785	0.634	0.674	0.766	0.757	0.545	0.662	0.752	6.35	3,142,119	42.2	8.3
✓	✓	✓	0.803	0.665	0.704	0.776	0.773	0.563	0.689	0.759	6.00	2,991,631	43.1	7.6

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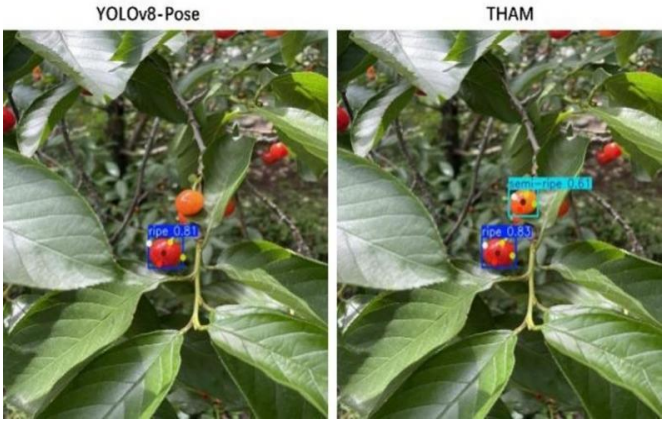


FIGURE 9. Visualization results of THAM ablation

On the left side of Fig. 9 are the detection results of the original YOLOv8-Pose, which accurately identifies "ripe" with a confidence score of 0.81. On the right side are the detection results of our model with THAM added, which accurately identifies "ripe" with a confidence score of 0.83 and "semi-ripe" with a confidence score of 0.61. The model with THAM added is able to detect smaller cherries due to its efficient detail perception and adaptive attention regulation capability. The 1×1 convolution reduces computational cost without losing key information; the special convolution and channel processing in the keypoint branch amplify the weak features of distant small targets. Furthermore, it can dynamically allocate attention, focus on the target, eliminate interference, and ensure un-interrupted feature propagation, enabling accurate localization of small and distant tar-gets even in complex environments.

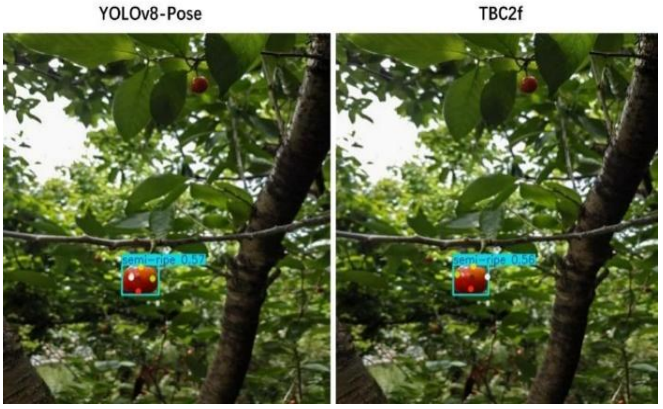


FIGURE 10. Visualization results of TBC2f ablation

On the left side of Fig. 10 are the detection results of the original YOLOv8-Pose, which accurately identifies "semi-ripe" with a confidence score of 0.57. On the right side are the detection results of our model with TBC2f added, which accurately identifies "semi-ripe" with a confidence score of 0.56. However, in terms of keypoint detection, YOLOv8-Pose in the left image does not achieve accurate recognition, whereas our model with TBC2f added in the right image accurately identifies all five keypoints. This is because TBC2f designs a dedicated enhancement branch for the keypoint task, extracting fine-grained local features through depthwise separable convolution and enhancing key feature responses with the SE module. At the same time, through parallel two-task design and a dynamic gating mechanism, it avoids interference from global features of the detection task, balances local details with global context, and relies on the C2f base structure to ensure feature quality, thereby achieving efficient and accurate keypoint localization.

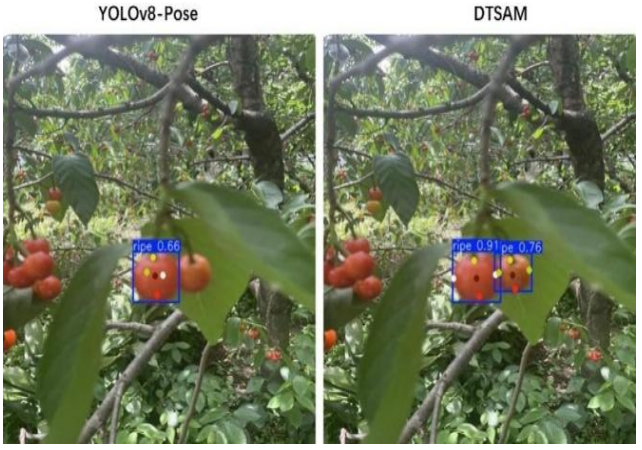


FIGURE 11. Visualization results of DTSAM ablation

On the left side of Fig. 11 are the detection results of the original YOLOv8-Pose, which identifies "ripe" with a confidence score of 0.66, but its keypoint recognition is not sufficiently accurate. On the right side are the detection results of our model with DTSAM added. Our model identifies two cherries of the "ripe" category with confidence scores of 0.91 and 0.76, respectively, and the keypoint detection is more accurate. This is because DTSAM, at the detection head level, exploits complementary information between the detection and keypoint tasks through explicit collaborative learning. Combined with task-specific feature enhancement and dynamic weight adjustment, it achieves simultaneous accuracy improvement for both tasks. It first provides a shared prior through a lightweight base attention layer, then uses dilated convolution to expand the receptive field and enhance spatial perception for the detection task, while employing depthwise separable convolution and multi-scale structures to enhance detail perception for the key-point task. Relying on a collaborative gate to dynamically adjust the participation weights of the two tasks, it concatenates the dual-task features via a collaborative learner, models complementary relationships, and enables knowledge transfer. Finally, through gated fusion, collaborative modulation, and multi-feature weighted fusion with learnable weights, it preserves the integrity of the original features, allowing weak-response features of a single task to be enhanced through collaboration. This fully leverages the exclusive advantages of both tasks and compensates for feature deficiencies, significantly improving

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object detection and keypoint localization accuracy under complex cherry images.

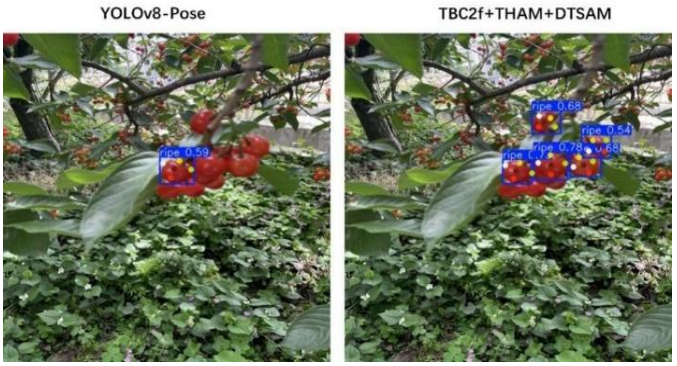


FIGURE 12. Visualization results of ablation with TBC2f, THAM, and DTSAM combined

On the left side of Fig. 12 are the detection results of the original YOLOv8-Pose, which identifies "ripe" with a confidence score of 0.59. On the right side are the detection results of our model with TBC2f, THAM, and DTSAM added. Our model identifies five cherries of the "ripe" category with confidence scores of 0.68, 0.76, 0.78, 0.68, and 0.54, respectively, and the keypoint positions are accurate. This is because our model, through a multi-level collaborative enhancement module, significantly improves the recognition capability for dense cherries. THAM dynamically allocates task attention at the deep feature layer, adapting to issues of occlusion and blurring in dense scenes. TBC2f balances the dual-task features during the feature extraction stage, enhancing the capture of contours and details of dense small targets. DTSAM explicitly exploits complementary task information at the detection head level, expands the detection receptive field, enhances detail perception, and combines dynamic weight fusion to strengthen weak-response features. Together, these three modules form a multi-scale collaboration chain, enabling the network to accurately distinguish individual cherries in dense clusters and capture subtle features, thereby identifying more cherries.

F. BOARD-LEVEL DEPLOYMENT EXPERIMENT ON NVIDIA JETSON ORIN NANO

TABLE 6. FPS data of different models on the NVIDIA Jetson Orin Nano

Model	YOLOv8-Pose	YOLOv11-Pose	YOLOv12-Pose	Ours
FPS	6.4	7.8	7.1	10.5
CPU utilization	37.1%	43.9%	44.7%	43.7%
GPU utilization	40.7%	44.3%	52.6%	43.8%
average power consumption	6.77w	7.24w	7.42w	6.93w

To quantitatively validate the practical deployability of our model on resource-constrained embedded devices, we conducted comprehensive inference performance tests on the NVIDIA Jetson Orin Nano platform, evaluating inference speed (FPS), CPU utilization, GPU utilization, and average power consumption, with all results summarized in Table 6. In terms of real-time performance, our model achieves a significant advantage, reaching 10.5 FPS, which outperforms YOLOv8-Pose, YOLOv11-Pose, and YOLOv12-Pose by 4.1, 2.7, and 3.4 FPS, respectively, effectively alleviating the inference bottleneck commonly encountered in embedded deployment scenarios. This speed gain, however, comes with a trade-off in resource efficiency: compared with the baseline YOLOv8-Pose—which exhibits the lowest FPS among all models—our model incurs moderately higher CPU utilization, GPU utilization, and average power consumption (increased by 0.16 W), representing a necessary compromise to achieve the substantial FPS improvement. By contrast, when compared with YOLOv11-Pose and YOLOv12-Pose, our model demonstrates clear overall superiority. Specifically, although YOLOv11-Pose shows slightly lower CPU utilization, our model achieves a 2.7 FPS advantage while maintaining lower GPU utilization and 0.31 W lower power consumption. Notably, YOLOv12-Pose attempts to boost accuracy through increased computational overhead, resulting in significantly higher GPU utilization and power consumption, yet its FPS still lags behind our model by 3.4 points—indicating an unfavorable trade-off where excessive resource consumption fails to deliver proportional speed gains. In summary, while our model does not achieve the ultra-low power consumption of YOLOv8-Pose, it delivers a substantial speed breakthrough, and it outperforms YOLOv11-Pose and YOLOv12-Pose in both GPU utilization and power efficiency. Overall, our model strikes a more favorable balance between

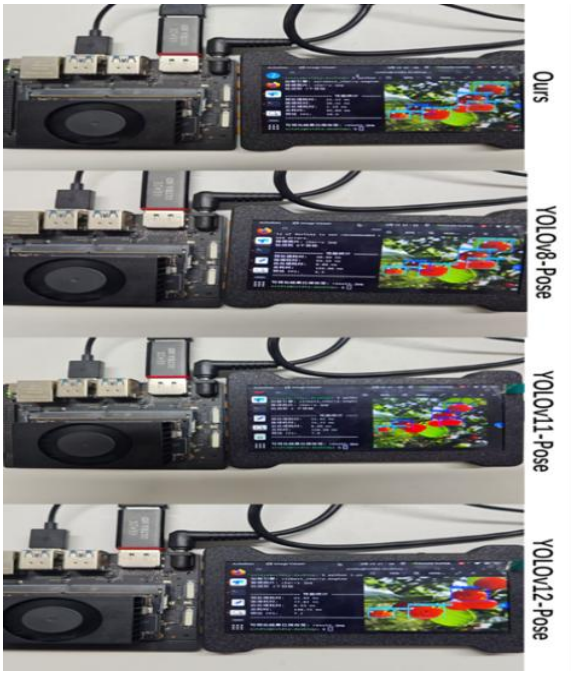


FIGURE 13. Comparison of detection performance of different models on the NVIDIA Jetson Nano.

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inference speed and resource consumption, demonstrating strong potential for high-real-time, cost-effective deployment on embedded devices in real agricultural scenarios.

G. EXPERIMENTAL ANALYSIS

This experiment fully validates the comprehensive performance of our proposed model in cherry detection and keypoint detection tasks. Overall, our model achieves a near-optimal balance among detection accuracy, processing speed, and model light-weightness, demonstrating strong practical value. Whether in bounding box localization accuracy (Detection mAP@0.5) or keypoint estimation accuracy (Keypoint mAP@0.5), our model significantly surpasses mainstream state-of-the-art models including YOLOv5 through YOLOv12. It is particularly worth emphasizing that our innovative fusion of the keypoint detection task with object detection not only imposes no performance burden but, through task synergy and information complementarity, further improves the overall detection accuracy, proving the effectiveness of the multi-task learning framework. Through the introduction of innovative modular designs, such as THAM for shallow feature enhancement, TBC2f for deep task balancing, and DTSAM for multi-scale synergy, the model achieves more efficient feature utilization. These modules work together to significantly enhance the model's representational capacity while effectively controlling its parameters and GFLOPs. Ultimately, while maintaining high accuracy, our model achieves a smaller model size and faster inference speed, fully meeting the stringent real-time requirements of actual orchard environments. In summary, with its high accuracy, high efficiency, and lightweight characteristics, our model provides a promising solution for intelligent fruit detection in complex agricultural scenarios.

V. Conclusion

In this paper, to address the practical requirements of multi-task visual perception for cherries in complex orchard environments, we propose YOLOv8-Cherry, a multi-level task collaborative enhancement network improved from YOLOv8-Pose. This network innovatively designs three collaborative modules: the Dual-task Balanced C2f module (TBC2f), the Task Harmonic Attention Module (THAM), and the Dual-Task Synergistic Attention Module (DTSAM), forming a complete collaborative system spanning feature extraction, semantic enhancement, and detection head optimization. Through TBC2f achieving dynamic balance between detection and keypoint tasks in shallow features, THAM performing adaptive attention allocation in deep semantics, and DTSAM promoting complementary information fusion between tasks at the prediction level, the system effectively resolves the issue of multi-task feature competition and conflict commonly found in traditional methods.

Experiments on the self-built cherry dataset demonstrate that YOLOv8-Cherry achieves significant improvements in detection and keypoint localization accuracy while maintaining real-time inference capability (43.1 FPS): the detection mAP@0.5 reaches 0.803, and the keypoint mAP@0.5 reaches 0.773, representing increases of 6.6% and 6.9%, respectively, over the baseline model. Ablation experiments further confirm the effectiveness of each module and their synergistic effects, with the model exhibiting excellent robustness particularly in complex scenarios involving severe occlusion, dense targets, and illumination variations. Furthermore, the model achieves a good balance between parameter efficiency and computational overhead, providing feasibility for edge device deployment.

The main contributions of this study include: 1. proposing a hierarchical task collaborative network architecture that achieves end-to-end joint optimization of cherry ripeness classification and grasping keypoint localization; 2. designing a lightweight task interaction mechanism and attention modulation strategy that effectively alleviates the feature competition problem in multi-task learning; 3. constructing and open-sourcing a cherry multi-task dataset containing multiple ripeness levels and various occlusion conditions, providing a benchmark for future research.

However, this study can be further explored in the following directions. First, YOLOv8-Cherry can be extended to more fruit categories (e.g., grapes, apples, etc.) to verify its cross-crop generalization capability. Second, temporal consistency optimization of the model in dynamic video streams can be investigated to enhance the perceptual stability of picking robots in mobile scenarios. Third, further lightweighting of the network architecture and hardware adaptation can be explored to facilitate its practical deployment in embedded platforms and agricultural robotic systems. Finally, by integrating motion planning and control of robotic arms, an intelligent picking system integrating "perception-decision-execution" can be constructed to achieve a closed-loop application from visual perception to precise grasping, further promoting the development of smart agriculture.

In summary, YOLOv8-Cherry provides an efficient and reliable visual perception solution for fruit detection and grasping pose estimation in complex agricultural environments. Its multi-level task collaborative design concept also offers a valuable reference for other agricultural multi-task vision systems.

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Qiaochu Li received his Master’s degree from Sichuan Agricultural University. His research interests focus on computer vision and deep learning, including image recognition, object detection, and related applications. He is currently dedicated to exploring efficient deep learning algorithms and practices for agricultural vision tasks.



Junyuan. Zhang current research interests include artificial intelligence, deep learning, and visual object detection for intelligent agricultural perception. He has published one peer-reviewed journal paper as the third author in Applied Sciences-Basel, focusing on advanced object detection algorithms and practical visual detection applications.



Yingnan Zhou is currently pursuing the bachelor’s degree in computer science and technology at the College of Engineering and Technology, Chengdu University of Technology. Her research focuses on computer vision and deep learning-based object detection. She has published one journal paper related to object detection as the second author in Applied Sciences-Basel.



Chen Zhang was born in Sichuan, China, in 2007. He is currently pursuing the bachelor’s degree in electronic science and technology at the College of Engineering and Technology, Chengdu University of Technology, Sichuan, China. His research interests include deep learning, keypoint detection and object detection based on computer vision.



Zhengzhou Li received his B.S. degree from Northeastern University, his M.S. and Ph.D. degrees from the Institute of Optics and Electronics, Chinese Academy of Sciences, and completed his postdoctoral research at Harvard University, USA. He is currently a Level-III Professor, doctoral and master’s supervisor, and postdoctoral cooperative supervisor, serving as a Chongqing academic and technical leader in information and communication engineering, an academic committee member of the Sichuan Key Laboratory of Aerospace Multi-source Information Fusion, a visiting researcher at two CAS institutes, and an expert in infrared seeker intelligent evaluation. His research interests mainly include infrared/visible detector technology, radar and optoelectronic signal processing, weak target detection and tracking, multi-source information fusion, and high-speed real-time signal processing systems. He has presided over a series of key scientific research projects such as the National Key R&D Program and the National Natural Science Foundation of China, has published more than 120 academic papers in journals including IEEE Transactions on Aerospace and Electronic Systems, and has obtained a first-class provincial and ministerial science and technology progress award. He also undertakes undergraduate and postgraduate teaching and training work and authored the textbook MATLAB Digital Signal Processing and Applications.



Jianguo Chen was born in Mengzhou, Henan, China, in 1978. He received the M.S. degree in communication and information systems from Harbin Engineering University in March 2005. He once engaged in the research of radio frequency and radiowave propagation at the China Research Institute of Radiowave Propagation. He is currently an Associate Professor and the program leader of intelligent science and technology at the College of Engineering and Technology, Chengdu University of Technology. He has long taught core professional courses, including Signals and Systems, Digital Signal Processing, and Digital Image Processing. His main research interests include radio frequency signal processing, chaotic ultra-wideband systems, and computer vision applications.

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